Research Evolution: Edition A to Edition B

FIN 43900 | Archer-Daniels-Midland Company (ADM) | September 1–October 16, 2026

Overall evolution. My original Edition A was an early investment thesis based mainly on historical financial performance. By Edition B, I had built a linked financial model and tested whether ADM's projected operating results, cash requirements, and valuation could justify buying the stock. My recommendation remained **WATCH/DEFER**, but it changed from a preliminary concern about declining profitability into a more specific, evidence-based investment decision.

1 | Where I started: Edition A

In Edition A, dated September 1, 2026, I focused on three trends in ADM's 2023–2025 results: revenue fell from **$93,935 million to $80,269 million**, gross margin declined from **8.00% to 6.27%**, and net income decreased from **$3,483 million to $1,078 million**. These results made me hesitant to recommend buying ADM immediately. I identified revenue growth, margin recovery, capital expenditures, inventory, and cash flow as questions to investigate, but I did not yet have a completed valuation or an estimated value per share.

2 | What changed in the research process

Dimension	Edition A: initial research	Edition B: completed analysis
Investment decision	Preliminary "Watch" based on deteriorating historical performance	WATCH/DEFER supported by valuation, scenario, and funding evidence
Financial evidence	Historical revenue, gross margin, and net income trends	2023–2025 inputs plus linked 2026–2030 income statement, balance sheet, and cash-flow forecasts
Valuation	No completed share-price estimate	FCFF DCF, WACC, terminal value, enterprise-to-equity bridge, and peer multiples
Business-specific risk	Inventory and spending identified as concerns	Inventory days, working capital, CapEx, and financing-facility limits modeled and stressed
Confidence and checks	Open questions and a research plan	Validation scripts, changed-input prediction, model checks, and cold-run documentation

3 | How the model changed my understanding

The biggest development was moving from asking whether ADM's earnings might recover to asking whether that recovery would produce enough **free cash flow** to support the market price. I built a five-year, three-statement pro-forma and linked operating assumptions to FCFF and a discounted cash flow valuation. This made it possible to see that revenue growth alone was not enough: higher inventory and other working-capital requirements, capital expenditures, and financing needs could absorb cash even when operating performance improved.

For the final October 16 analysis, the Base assumptions included **2.0% annual revenue growth, a 6.75% gross margin, 51 inventory days, and $1,400 million of annual CapEx**. The model also used a $500 million minimum-cash policy and a $6,000 million financing-facility limit. Those inputs turned the broad risks I mentioned in Edition A into measurable drivers and constraints.

4 | How the final valuation affected my decision

At the October 16, 2026 observed close of **$82.61 per share**, my final linked DCF produced a **Base mathematical residual equity value of approximately –$0.76 per share** and a **funded High scenario of approximately $24.22 per share**. The Low scenario produced approximately –$14.32 per share but exceeded the $6,000 million facility limit, so I treated it as a diagnostic stress test rather than an investable valuation. Negative residual values are outputs of the model's enterprise-to-equity bridge, **not forecasts that the stock will trade below zero**.

I also compared ADM with Bunge (BG) and Ingredion (INGR) using P/E and EV/EBITDA methods. One BG EV/EBITDA result implied approximately **$83.72 per share**, close to the market price. Instead of averaging this with the DCF, I examined why the methods disagreed, including differences in peer businesses, EBITDA definitions, capital structure, and the DCF's assumptions. That disagreement reinforced the need for a conditional recommendation rather than a single precise price target.

5 | Testing assumptions and checking the work

My final project went beyond a single base case. I ran low, base, and high scenarios; tested changes in operating assumptions and WACC/terminal growth; and used a facility-cap check to identify cases that could not be financed within the model's limits. I also documented validation work, including an ABG known-answer exercise, an ADM filing-input check, a locked prediction before a changed-input run, accounting and valuation checks, and a fresh-environment cold run. These steps helped distinguish a model output from a result I could reasonably defend.

6 | AI's role and my responsibility

I used ChatGPT and Codex to help develop and troubleshoot Python scripts, organize the model, create tests, and improve documentation. AI made it easier to move from an initial research plan to a working valuation application, but I still needed to decide which assumptions were reasonable, compare outputs with source evidence, investigate failures, and explain conflicting valuations. The final WATCH/DEFER recommendation is my investment judgment based on the documented evidence, not a recommendation accepted simply because AI produced it.

7 | What I learned and what could change my view

The main lesson from this project is that a company can look attractive because it is established or because one valuation multiple supports the market price, yet still fail to generate enough modeled cash flow under a reasonable set of assumptions. I started by noticing declining historical results; I finished by identifying the specific conditions that matter most for ADM: sustained margin improvement, stronger cash conversion, manageable reinvestment, and adequate financing capacity. I would revisit WATCH/DEFER if new reported operating results or better-supported assumptions materially improved those drivers and brought the linked valuation closer to the observed stock price.

Evidence and scope

Prepared from the project's published Edition A, final investment committee memo, decision analysis, and README. The document describes the submitted research evolution; it does not independently certify the model calculations or replace the original Edition A or other required evidence.

Edition A

Investment committee memo

Decision analysis